Chinmay Joshi

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Profile

I am a **machine learning–oriented data scientist** with experience across **applied ML**, **training, fine-tuning, and evaluating neural networks**, **analytics**, and **computer vision**, backed by a strong foundation in **statistics**. I am interested in building **data-centric, reliable solutions** with a practical impact.

Education

Universität des Saarlandes — Saarbrücken, Germany
Master's in Computer Science (GPA: 1.7) — *Oct 2022 – Mar 2026*

- Master's Thesis: **On the effect of Data Pruning on Memorization**
  (*Advisor: Dr. Rebekka Burkholz – CISPA Helmholz Center for Information Security*)
  - Investigating the impact of **Data Pruning** on **Memorization in Neural Networks**, with a comparative **empirical analysis of training behaviour** across different memorization definitions, with experiments ranging from CIFAR + Resnets to transfer learning of a pretrained ViT on Imagenet to Flowers and Pets.
  - Implemented **large-scale experiments** on A100s using SLURM, eg, estimating stability-based memorization score (Lukasik et al.) with 200 training runs per experiment.
- Recipient of **SaarlandStipendium** and **Santander Scholarship**.

Narsee Monjee Institute of Management Studies — Mumbai, India
Bachelors in Data Science (GPA: 3.8/4) — *Aug 2018 – May 2022*

Experience

Research Assistant — Dec 2025 – Present
Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) — *Kaiserslautern, Germany*

- Developing a modular **Subgroup Discovery engine** to identify statistically interesting patterns in static as well as **time-dependent datasets**. (Source Code)
- Engineering a hybrid approach combining **clustering-based candidate generation**, **KL divergence scoring**, and **interpretable rule-based explanations** (via Random Forests) to automatically detect and explain meaningful data anomalies.

Research Assistant — Mar 2025 – Sep 2025
Fraunhofer IZFP – Proj. Urbanist — *Saarbrücken, Germany*

- Analyzed **environmental and mobility sensor data** to generate actionable insights on how **transport choices affect environmental impact**, directly supporting sustainability-focused decision-making.
- Modeled raw sensor signals into **interpretable metrics** (UV index, sun exposure, noise levels) to enable real-time evaluation of route sustainability.
- Developed **geospatial routing algorithms** using **OpenStreetMap** to visualize environment-aware routes for a gamified urban mobility application.

Research Intern Apr 2024 – Aug 2024
National Institute of Informatics *Tokyo, Japan*

- Created a **synthetic 3D relighting dataset** using **Blender**, Objaverse assets, and PolyHaven environments; generating **150K+ multi-view, multi-light samples** with corresponding **depth maps and surface normals**.
- Explored **single-image 3D reconstruction with relighting** by conditioning environment lighting on a **diffusion-based prior (Zero-1-to-3)**, analyzing challenges in illumination and view consistency due to the ill-posed nature of the task.

Research Assistant Aug 2023 – Mar 2024
Fraunhofer IZFP – Proj. SmartPigHome *Saarbrücken, Germany*

- Built a lightweight **audio annotation tool** that replaced a manual paper-based workflow, allowing veterinarians to label data efficiently and **reducing labeling time by up to 80%**.
- Supported computer vision experiments involving **object detection and optical flow** for tracking livestock in video data.

Research Assistant Jun 2023 – Mar 2024
Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) – Proj. KIMonoS *Saarbrücken, Germany*

- Developed an **algorithmic scheduling system** for multi-vehicle shuttle routing, optimizing **passenger wait times and seat utilization** using Python.
- Evaluated system performance under a **12-hour, 60-passenger stochastic simulation** in SUMO to assess behavior under realistic demand patterns.

Computer Vision Intern Jan 2022 – Apr 2022
Aurify Systems Pvt. Ltd. *Mumbai, India*

- Developed **computer vision pipelines** for retail and warehouse analytics, including **lightweight CNN-based object detection** and **template matching with non-maxima suppression** for **bin detection and empty-shelf monitoring**; solutions were showcased in a public exhibition.
- Built a **pose-based activity detection system** using **AlphaPose** and a **classical ML classifier** (e.g., Random Forest) to detect suspicious activities, achieving **98% accuracy** in controlled field tests.
- Prototyped a **CCTV video analytics tool**, combining **image processing, pose estimation, and object detection** to identify workers using mobile phones through a **multi-stage decision pipeline**.

Projects and Publications

- **Algorithmic Trading Strategy** (IEEE I2CT 2022): Presented **PCA + LSTM** for **forecasting trade signals** using a multicollinear set of 84 technical indicators as features, and leveraging **dimension reduction** for an improved **Time-Series** modeling. Complemented with portfolio optimization strategies, we attain **20% more returns than the Buy-and-Hold benchmark**. (Paper) (Source Code)
- **Hierarchical VAE for Image Compression**: Implemented a **hierarchical variant of an existing VAE-based image compression model** (Balle et al.), achieving **+5 PSNR over JPEG** at similar bitrate through architectural modifications. (Report)

Skills

Programming Languages: Python, SQL, R

Tooling & Frameworks: PyTorch, Tensorflow, Hugging Face, Tableau, Weights & Biases, Git (Version Control), Docker, OpenStreetMap, SLURM, FastAPI, Langchain

Conceptual Understanding: CNNs, LSTMs, Transformers, Diffusion Models, Large Language Models (LLMs), RAGs, Causal Inference, Exploratory Data Analysis, Pattern Mining, Signal Processing, Camera Geometry, 3D Computer Vision, Image Analysis, Time-Series Analysis, Statistics

Languages: English (Native), German (A2), Hindi (Native), Marathi (Native)